Zhe (Jay) Xu

Research Scientist @ Meta
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Education

University of Illinois Urbana-Champaign — Champaign, IL, USA
Ph.D. in Computer Science Thesis: Optimal graph learning — *May 2021 – April. 2025*

University of Illinois Urbana-Champaign — Champaign, IL, USA
M.S. in Computer Science Thesis: Dense subgraph detection on multi-layered networks — *Aug. 2019 – May 2021*

Fudan University — Shanghai, China
B.E. in Electronic Engineering — *Sep. 2014 – Jun. 2018*

Publication

Preprint Papers

P1. **Zhe Xu**, et al. The Case Against Generation for Retrieval: Discriminative Language Models as Effective Retrievers. Under review.

P2. Xinyu Zhu, **Zhe Xu**, Xiaohan Wei, Yunchen Pu, Fei Tian, Chonglin Sun, Kaushik Rangadurai, et al. Self-Guided Test-Time Training for Long-Context LLMs. Under review.

P3. Yufei Li, Zaiwei Zhang, Mingfu Liang, Kavosh Asadi, **Jay Xu**, Jimmy Kim, Chongyang Bai, Jieyi Zhang, et al. GR2 Technical Report. Under review.

P4. Mingfu Liang*, Yufei Li*, **Jay Xu***, Kavosh Asadi, Xi Liu, et al. Generative Reasoning Re-ranker. Under review.

P5. Beidi Zhao, Boxin Du, **Zhe Xu**, Liangyue Li, Hanghang Tong. Learning Optimal Propagation for Graph Neural Networks. Under review.

P6. Yuchen Yan, Yuzhong Chen, Huiyuan Chen, Xiaoting Li, **Zhe Xu**, Mahashweta Das, Hanghang Tong. THiGCN: Temporal Heterophilic Graph Convolutional Network for Event-based Continuous Graphs. Under review.

Accepted Conference Papers

C1. **Zhe Xu**, Ankit Peshin, Chiyu Zhang, Feng Qi, Johnson Lui, Anil Ramakrishna, et al. Tokenizing Numerical and Embedding Features for LLM RecSys. TokShop at COLM 2026.

C2. **Zhe Xu**, Kaveh Hassani, Si Zhang, Hanqing Zeng, Michihiro Yasunaga, Limei Wang, Dongqi Fu, Ning Yao, Bo Long, Hanghang Tong. Language Models are Graph Learners. Findings of EACL 2026.

C3. **Zhe Xu**, Menghai Pan, Yuzhong Chen, Huiyuan Chen, Yuchen Yan, Mahashweta Das, Hanghang Tong. Fine-grained Graph Rationalization. CIKM 2025.

C4. **Zhe Xu**, Ruizhong Qiu, Yuzhong Chen, Huiyuan Chen, Xiran Fan, Menghai Pan, Zhichen Zeng, Mahashweta Das, Hanghang Tong. Discrete-state Continuous-time Diffusion for Graph Generation. NeurIPS 2024.

C5. **Zhe Xu**, Yuzhong Chen, Menghai Pan, Huiyuan Chen, Mahashweta Das, Hao Yang, Hanghang Tong. Kernel Ridge Regression-Based Graph Dataset Distillation. SIGKDD 2023.

C6. **Zhe Xu**, Yuzhong Chen, Qinghai Zhou, Yuhang Wu, Menghai Pan, Hao Yang, Hanghang Tong. Node Classification Beyond Homophily: Towards a General Solution. SIGKDD 2023.

C7. **Zhe Xu**, Kaize Ding, Yu-Xiong Wang, Huan Liu, Hanghang Tong. Generalized Few-Shot Node Classification on Graphs. ICDM 2022. **(Best-Ranked Papers)**.

C8. **Zhe Xu**, Boxin Du, Hanghang Tong. Graph Sanitation with Application to Node Classification. TheWebConf 2022.

C9. **Zhe Xu**, Si Zhang, Yinglong Xia, Liang Xiong, Jiejun Xu, Hanghang Tong. DESTINE: Dense Subgraph Detection on Multi-Layered Networks. CIKM 2021.

C10. **Zhe Xu**, Si Zhang, Yinglong Xia, Liang Xiong, Hanghang Tong. Ranking on Network of Heterogeneous Information Networks. IEEE BigData 2020.

C11. Dongqi Fu*, **Zhe Xu***, Bo Li, Hanghang Tong, and Jingrui He. A View-Adversarial Framework for Multi-View Network Embedding. CIKM 2020. (*equal contribution)

C12. Ruizhong Qiu, **Zhe Xu**, Wenxuan Bao, Hanghang Tong. Turing Completeness of Prompting. ICLR 2025.

C13. Huiyuan Chen, **Zhe Xu**, Chin-Chia Michael Yeh, Vivian Lai, Yan Zheng, Minghua Xu, Hanghang Tong. Masked Graph Transformer for Large-Scale Recommendation. SIGIR 2024.

C14. Qinghai Zhou, Yuzhong Chen, **Zhe Xu**, Yuhang Wu, Menghai Pan, Mahashweta Das, Hao Yang, Hanghang Tong. Graph Anomaly Detection with Adaptive Node Mixup. CIKM 2024.

C15. Zhichen Zeng, Ruizhong Qiu, **Zhe Xu**, Zhining Liu, Yuchen Yan, Tianxin Wei, Lei Ying, Jingrui He, Hanghang Tong. Graph Mixup on Approximate Gromov–Wasserstein Geodesics. ICML 2024.

C16. Haiyun Jiang, Li Cui, **Zhe Xu**, Deqing Yang, et al. Relation Extraction Using Supervision from Topic Knowledge of Relation Labels. IJCAI 2019.

C17. Dongqi Fu*, Liri Fang*, Zihao Li*, **Zhe Xu***, Hanghang Tong, Vetle I Torvik, Jingrui He. Bring Complex Geometric Information to LLMs: A Positional Survey of Graph Parametric Representation. LOG 2025. (*equal contribution)

C18. Haobo Xu, Yuchen Yan, Dingsu Wang, **Zhe Xu**, Zhichen Zeng, Tarek F. Abdelzaher, Jiawei Han, Hanghang Tong. SLOG: An Inductive Spectral Graph Neural Network Beyond Polynomial Filter. ICML 2024.

C19. Xinyu He, Chenhan Xiao, Haoran Li, Ruizhong Qiu, **Zhe Xu**, Yang Weng, Jingrui He, Hanghang Tong. PowerGrow: Feasible Co-Growth of Structures and Dynamics for Power Grid Synthesis. SIGKDD 2026.

C20. Zhining Liu, Ruizhong Qiu, Zhichen Zeng, Hyunsik Yoo, David Zhou, **Zhe Xu**, Yada Zhu, Kommy Weldemariam, Jingrui He, Hanghang Tong. Class-Imbalanced Graph Learning without Class Rebalancing. ICML 2024.

Accepted Journal Papers

J1. **Zhe Xu**, Kaize Ding, Yu-Xiong Wang, Huan Liu, Hanghang Tong. Generalized Few-Shot Node Classification on Graphs: Toward an Uncertainty-Based Solution. KAIS 2023.

J2. Kaize Ding, **Zhe Xu**, Hanghang Tong, Huan Liu. Data Augmentation for Deep Graph Learning: A Survey. SIGKDD Explorations, 2022.

J3. Qing Chen, Nan Chen, Guande Wu, Ziyan Liu, **Zhe Xu**, Hanghang Tong, Nan Cao. Calliope-Net: Automatic Generation of Graph Data Facts via Annotated Node-link Diagrams. IEEE Transactions on Visualization and Computer Graphics 2023.

Talks

T1. *Lecture-Style Tutorial*: Natural and Artificial Dynamics in GNNs: A Tutorial. WSDM 2023.

T2. *Invited Talk*: Graph Data Distillation. Visa Research. July 2023.

Research Experience

Research Scientist — Sunnyvale, CA, USA
Monetization, Meta. Manager: Dr. Wen-Yen Chen. — *Mar. 2025 – Present*
- Contribute to the ads recommender system at Meta.

Research Intern — Sunnyvale, CA, USA
Monetization, Meta. Manager: Dr. Si Zhang. — *May. 2024 – Nov. 2024*
- Research on the interaction between large language models and graph machine learning.

PhD Intern — Palo Alto, CA, USA
Visa Research, Visa Inc. Manager: Dr. Menghai Pan. — *May. 2023 – Aug. 2023*
- Conducted research on graph data augmentation in the life cycle of graph machine learning.

Research Intern New York City, NY, USA
Applied Machine Learning Team, Meta. Manager: Kelvin Niu. *May. 2022 – Aug. 2022*

- Constructed a large-scale representation learning system on user-video interaction data via graph neural networks.

Algorithm Engineer Intern Hangzhou, Zhejiang, China
Search Department, Alibaba Group. Manager: Dr. Muhua Zhu. *Jun. 2018 – Aug. 2018*

- Constructed E-commerce knowledge graphs on specific shopping topics.

Graduate Research Assistant Urbana, IL, USA
University of Illinois Urbana-Champaign. Advisor: Prof. Hanghang Tong. *Aug. 2019 – Present*

- Conducted research on augmenting graph data

Graduate Research Assistant Tempe, AZ, USA
Arizona State University. Advisor: Prof. Hanghang Tong. *Aug. 2018 – Aug. 2019*

- Conducted research on multi-layered networks.

Undergraduate Research Assistant Shanghai, China
Fudan University. Advisor: Prof. Yanghua Xiao. *Sep. 2016 – Jun. 2018*

- Constructed a financial product knowledge base and a financial product Q&A system (Chinese).

Grant Experience

- Helped a successful NSF proposal on 'SLES: NetSafe: Towards a Computational Foundation of Safe Graph Neural Networks.'
- Helped a successful NSF proposal on 'Collaborative Research: Towards a Theoretic Foundation for Optimal Deep Graph Learning.'

Teaching Experience

Teaching Assistant Champaign, IL, USA
CS 514 Advanced Topics in Network Science *Fall 2023*

Teaching Assistant Champaign, IL, USA
CS 512 Data Mining Principle *Fall 2022, Spring 2021*

Teaching Assistant Champaign, IL, USA
CS 412 Introduction to Data Mining *Fall 2021*

Service

Conference reviewer: AISTATS (2025), ICML (2024), SDM (2024), ICLR (2024-2025), AAAI (2023-2025), NeurIPS (2023-2024), KDD (2023,2025), CIKM (2021-2025), TheWebConf (2023-2025).
Journal reviewer: TNNLS, TVCG, TKDD.
Workshop reviewer: GLFrontiers (2024), TrustLOG (2022)
Conference subreviewer: ICDCS (2022), SIGIR (2021), CIKM (2019), ICDCS (2019).